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batch-26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab																		
Academic Year: 2025-2026																				
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M. Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M. Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
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Course Code	23CS002PC304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week 1 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52																	
Duration	2 Hours	Applicable to Batches	All batches																	
Assignment Number: 1.3 (Present assignment number) / 24 (Total number of assignments)																				
	Q.No.	Question	Expected Time to complete																	

Lab 2: Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI

Lab Objectives:

- To explore and evaluate the functionality of Google Gemini for AI-assisted coding within Google Colab.
- To understand and use Cursor AI for code generation, explanation, and refactoring.
- To compare outputs and usability between Gemini, GitHub Copilot, and Cursor AI.
- To perform code optimization and documentation using AI tools.

Lab Outcomes (LOs):

After completing this lab, students will be able to:

- Generate Python code using Google Gemini in Google Colab.
- Analyze the effectiveness of code explanations and suggestions by Gemini.
- Set up and use Cursor AI for AI-powered coding assistance.
- Evaluate and refactor code using Cursor AI features.
- Compare AI tool behavior and code quality across different platforms.

Task 1: Word Frequency from Text File

- **Scenario:**
You are analyzing log files for keyword frequency.
- **Task:**
Use Gemini to generate Python code that reads a text file and counts word frequency, then explains the code.
- **Expected Output:**
 - Working code
 - Explanation
 - Screenshot

The screenshot displays the Cursor AI IDE interface. On the left, a sidebar titled 'Python word frequency counter' provides an explanation of the code. It states that the code reads a text file at a specified path, converts it to lowercase, extracts words using a regex pattern (letters, digits, and underscores), and counts their frequencies using the Counter class. It also lists small improvements, such as correcting the error message to match the actual file_path and adding a try-except block for file handling.

The main editor shows the following Python code:

```
1 from collections import Counter
2 import re
3
4 def count_word_frequency(file_path):
5     try:
6         with open(file_path, 'r') as file:
7             # Read file and convert to lowercase
8             text = file.read().lower()
9             # Use regex to find words (ignores punctuation)
10            words = re.findall(r'\w+', text)
11
12            # Count frequencies
13            word_counts = Counter[Any](words)
14
15            print(f"{'Word':<15} | {'Frequency':>10}")
16            print("-" * 25)
```

At the bottom, the terminal output shows the execution of the code, displaying the word frequency counts for a sample text file.